

temperature of Z is higher than 30°C.

14. Answer: C

For process 1, thermodynamic state X $\rightarrow$ thermodynamic state Y.

$$\begin{aligned}\Delta U &= Q + W \\ &= 2800 + (-600) \\ &= 2200 \text{ kJ}\end{aligned}$$

For process 2, thermodynamic state Y $\rightarrow$ thermodynamic state X.

For cyclic process, $\Delta U = 0$. Hence $\Delta U = -2200 \text{ kJ}$ for process 2.

$$\begin{aligned}\Delta U &= Q + W \\ -2200 &= 1000 + W \\ W &= -3200 \text{ kJ}\end{aligned}$$

Hence work done by the system during process 2 is 3200 kJ.